EXPERIMENT :19

Install a C compiler in the virtual machine created using virtual box / VM Ware Workstation / Player and execute Simple C Programs.

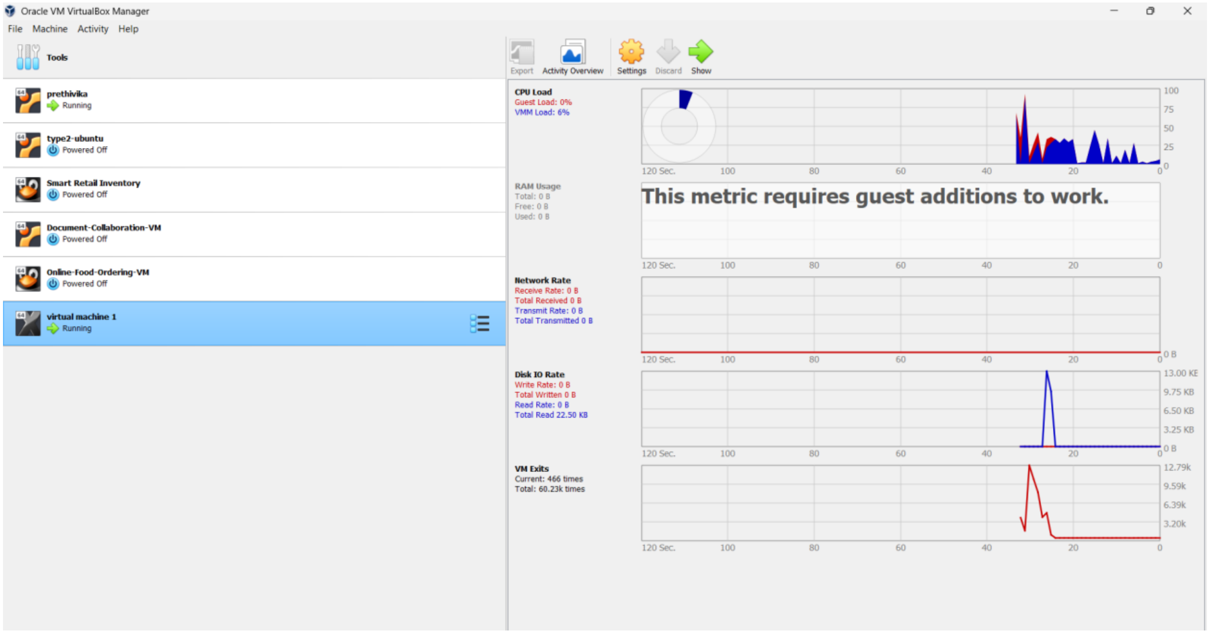
AIM:

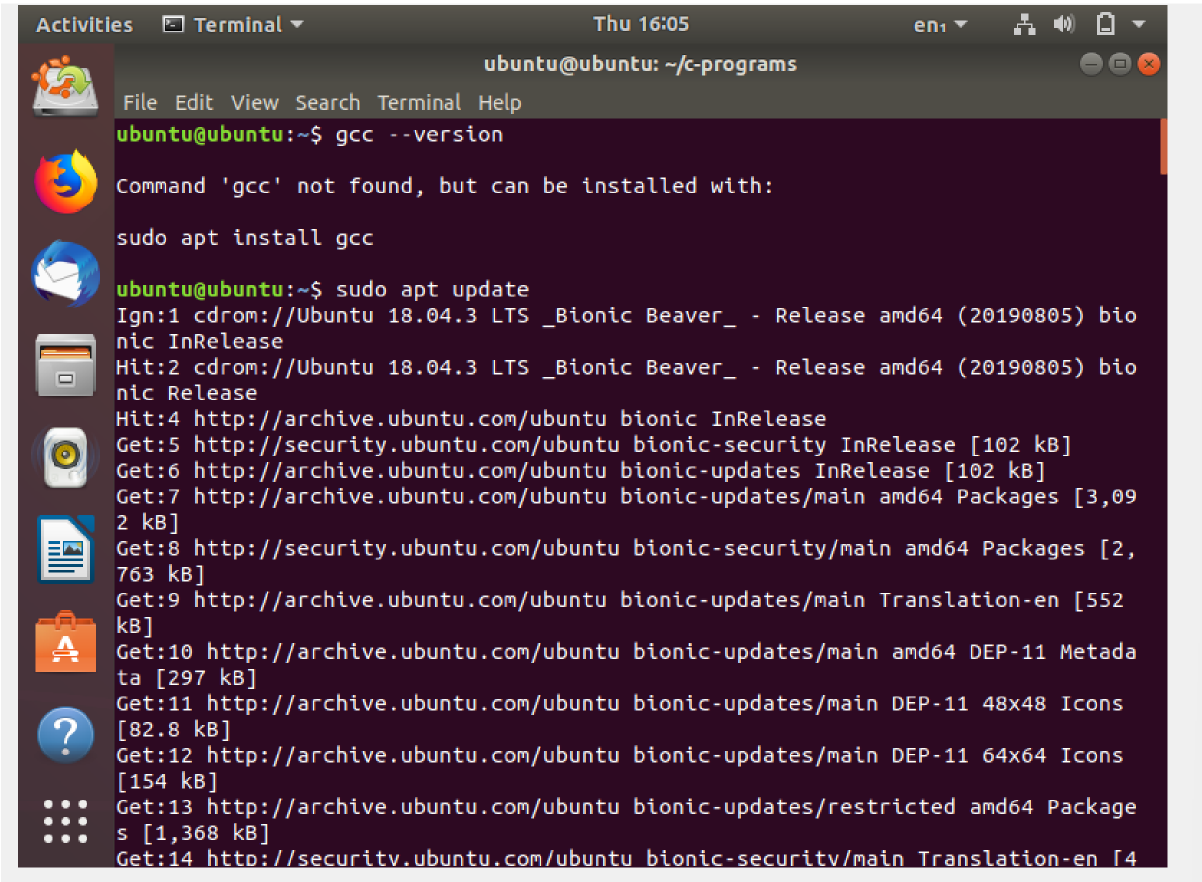
To install the GCC C compiler in an Ubuntu virtual machine created using Oracle VirtualBox and compile and execute a simple C program to determine whether a given number is even or odd.

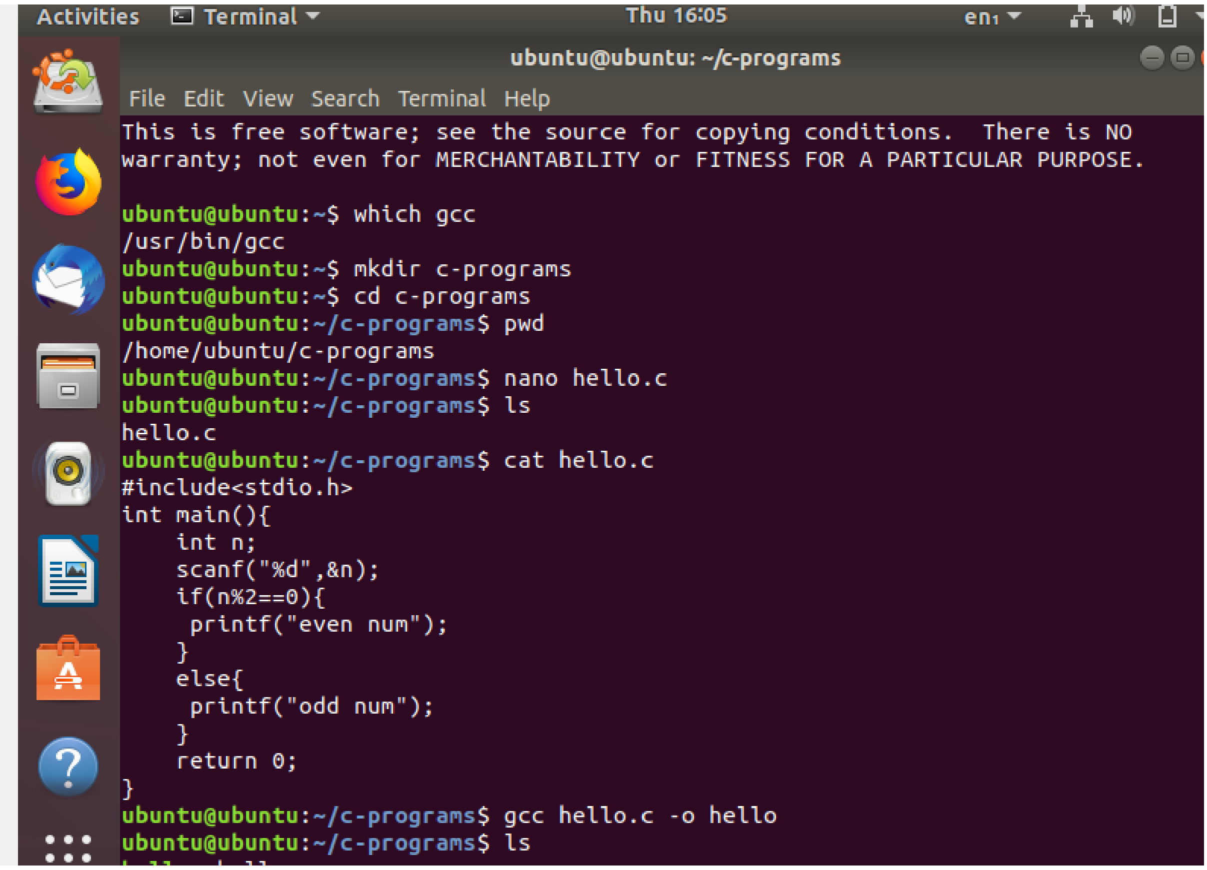
PROCEDURE:

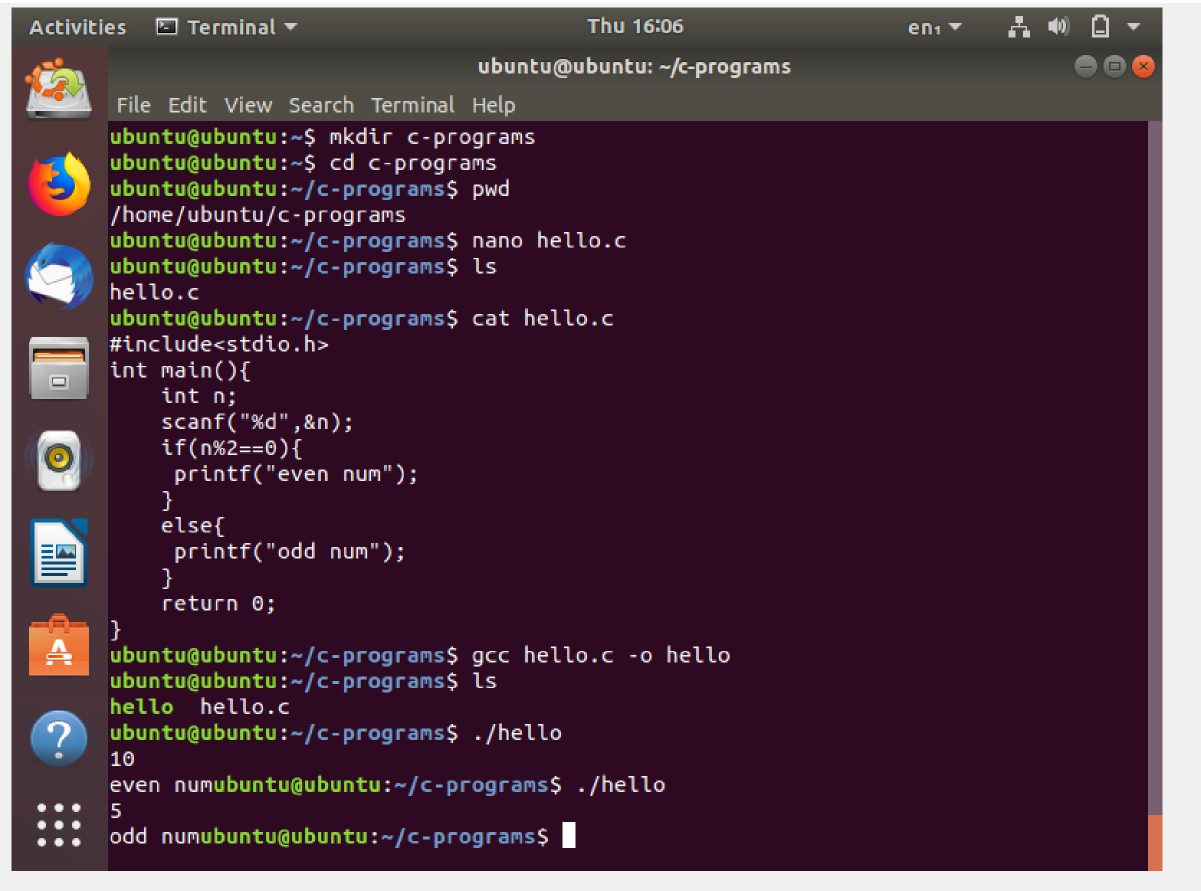
1. Open Oracle VirtualBox, start the existing Ubuntu Virtual Machine, and log in to the Ubuntu desktop.
2. Open the Terminal and update the package repository using sudo apt update.
3. Install the GCC C compiler and required development tools using sudo apt install build-essential.
4. Verify the compiler installation using gcc --version and confirm that GCC is available in the Ubuntu virtual machine.
5. Create a C source file using a text editor such as Nano and enter the Even and Odd Number program using the modulus (%) operator.
6. Compile the program using gcc filename.c -o filename and execute it using ./filename, then enter a number when prompted.
7. Verify the displayed output and confirm whether the entered number is Even or Odd, demonstrating successful C program compilation and execution inside the Ubuntu virtual machine.

IMPLEMENTATION:






Activities
Terminal
Thu 16:05
en₁
ubuntu@ubuntu: ~/c-programs
File Edit View Search Terminal Help
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
ubuntu@ubuntu:~$ which gcc
/usr/bin/gcc
ubuntu@ubuntu:~$ mkdir c-programs
ubuntu@ubuntu:~$ cd c-programs
ubuntu@ubuntu:~/c-programs$ pwd
/home/ubuntu/c-programs
ubuntu@ubuntu:~/c-programs$ nano hello.c
ubuntu@ubuntu:~/c-programs$ ls
hello.c
ubuntu@ubuntu:~/c-programs$ cat hello.c
#include<stdio.h>
int main(){
int n;
scanf("%d",&n);
if(n%2==0){
printf("even num");
}
else{
printf("odd num");
}
return 0;
}
ubuntu@ubuntu:~/c-programs$ gcc hello.c -o hello
ubuntu@ubuntu:~/c-programs$ ls


Activities
Terminal
Thu 16:06
en₁
ubuntu@ubuntu: ~/c-programs
File Edit View Search Terminal Help
ubuntu@ubuntu:~$ mkdir c-programs
ubuntu@ubuntu:~$ cd c-programs
ubuntu@ubuntu:~/c-programs$ pwd
/home/ubuntu/c-programs
ubuntu@ubuntu:~/c-programs$ nano hello.c
ubuntu@ubuntu:~/c-programs$ ls
hello.c
ubuntu@ubuntu:~/c-programs$ cat hello.c
#include<stdio.h>
int main(){
int n;
scanf("%d",&n);
if(n%2==0){
printf("even num");
}
else{
printf("odd num");
}
return 0;
}
ubuntu@ubuntu:~/c-programs$ gcc hello.c -o hello
ubuntu@ubuntu:~/c-programs$ ls
hello hello.c
ubuntu@ubuntu:~/c-programs$ ./hello
10
even numubuntu@ubuntu:~/c-programs$ ./hello
5
odd numubuntu@ubuntu:~/c-programs$

RESULT:

The GCC C compiler was successfully installed in the Ubuntu virtual machine using Oracle VirtualBox. The Even and Odd Number C program was successfully compiled and executed, and the program correctly identified the given number as either **even or odd**.